# Deepak Mallampati

## APPLIED AI ENGINEER

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Kent, OH | US-based, F-1 OPT, open to sponsorship

## PROFESSIONAL SUMMARY

Applied AI Engineer with 4+ years delivering end-to-end AI systems in regulated environments - agentic LLM platforms at Progress Solutions (healthcare RAG, 42% token reduction, RAGAS/BERTScore evaluation) and Vanguard (25+ AI agents, 100,000+ daily requests, 42% unsafe-output reduction). Strong fit for Scale AI's Global Public Sector Applied AI team: comfortable partnering with stakeholders to define AI solutions, building production-grade evaluation pipelines, and scaling LLM applications across teams. MS CS Kent State (2025). US-based, F-1 OPT - open to UK relocation with sponsorship.

## CORE COMPETENCIES

Applied AI / LLM Engineering • Client-Facing AI Delivery • LLM Evaluation Frameworks • RAG + Agentic Systems • Cross-Team AI Rollout • Regulated-Industry AI (Healthcare, Finance) • Python, TypeScript, Java • AWS, GCP, Azure

**Generative AI & LLMs:** GPT-4o, GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG Architecture | Prompt Engineering | Multi-Agent Systems | Fine-tuning (LoRA, QLoRA, PEFT) | Content Safety | Hugging Face Transformers
**Machine Learning & NLP:** PyTorch, TensorFlow, Keras | NER, Text Classification, Summarization | Model Optimization | A/B Testing | Drift Detection | RAGAS Evaluation | BERTScore
**Vector Search & Retrieval:** FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Document Chunking Strategies | Semantic Caching
**MLOps & Cloud Infrastructure:** AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure (OpenAI, AI Services, Content Safety), GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD (Jenkins, Azure DevOps) | MLflow, Weights & Biases | Model Monitoring & Observability
**Data Engineering:** PostgreSQL, MongoDB, Redis | Spark, Databricks | Kafka | Pandas, NumPy | Data Lineage & Governance
**Security & Compliance:** HIPAA, SOC 2, PCI DSS | OAuth 2.0, JWT, RBAC | PII Masking/Redaction | Prompt Injection Mitigation | Audit Logging | Zero-Trust Architecture
**Languages & Tools:** Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git, Jira, Confluence | Agile/Scrum

## PROFESSIONAL EXPERIENCE

**Progress Solutions** *Jul 2025 - Present*

*AI Engineer - USA*

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Designed and operationalized **privacy-preserving data workflows** for sensitive clinical datasets, implementing K-anonymity, L-diversity, and differential privacy (Laplace mechanism) to enable compliant analytics and model development with zero PII exposure.
- Built fault-tolerant **automation infrastructure** featuring scheduled batch orchestration and exponential-backoff retry logic, improving pipeline reliability and reducing failed production runs by **60%**.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) paired with latency and accuracy dashboards to benchmark model iterations pre-deployment and surface regressions before release.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

### Vanguard
*Jan 2024 - Jan 2025*

*AI Engineer Intern - USA*

- Developed and enhanced **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across **25+ AI agents** and workflows.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini** models, improving task success rates by **27%** on internal evaluation datasets and informing platform model-selection decisions.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

### Kent State University
*Oct 2023 - Jan 2024*

*ML & Gen AI Research Assistant - USA*

- Built a **privacy-preserving ML pipeline** on a clinical hospital-readmission dataset, applying k-anonymity (k=3), l-diversity (l=2), and Laplace differential privacy across four ε levels (0.1-2.0) over 8 clinical features.
- Reduced record-linkage re-identification risk from **3.38% to 0.32%** and eliminated unique-record exposure (0.94% to 0%) under k-anonymity + l-diversity, while retaining **99% of baseline predictive performance** (Random Forest AUC 0.689 vs. 0.696; weighted-F1 0.895).
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 using a curated 361-example instruction dataset, conditioning the model on a six-stage narrative schema for controllable long-form generation.
- Authored a peer-review-ready **IEEE-format conference paper** documenting the methodology, experiments, and results (LaTeX).

### Emerson
*Jun 2022 - Apr 2023*

*ML Trainee - India*

- Developed and trained supervised **regression and classification models** on operational pipeline-storage data, applying standard ML workflows to equipment-failure prediction, anomaly detection, and capacity forecasting.
- Fine-tuned **BERT and RoBERTa** models for entity extraction and text classification on domain-specific financial datasets, achieving **89% F1-score** on custom NER tasks.

## SELECTED PROJECTS

### MangaLens
A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga and webtoons across **29+ sites**, turning a multi-step manual translation chore into a single inline action.

### career-ops
An autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight - compressing hours of repetitive job-hunting into an automated workflow.

### Privacy-Preserving Clinical ML Pipeline
A framework that lets researchers analyze sensitive patient data without exposing identities, combining k-anonymity, l-diversity, and differential privacy - quantifies exactly how much privacy each technique buys for how little accuracy cost.

### Story Director LLM
A fine-tuned language model that generates structured short-form video narratives from a single prompt, paired with an automated rendering pipeline.

## EDUCATION

**M.S. in Computer Science** - *Kent State University, OH* *2025*

GPA: 3.70 / 4.0